Exoplanet Detection and Characterization Report

Team Vision

STAC Club, IIT Mandi

Mission: Operation Dim Star

NOTE: All the calculations are done with the help of Python , Pandas, Matplotlib and hence attaching code.

Abstract — This report details our calculations to find a new home for the Rebellion. We analyzed the brightness data of three stars (STAC-1, STAC-2, and STAC-3) to find planets, calculate their sizes and orbits, and choose the best candidate.

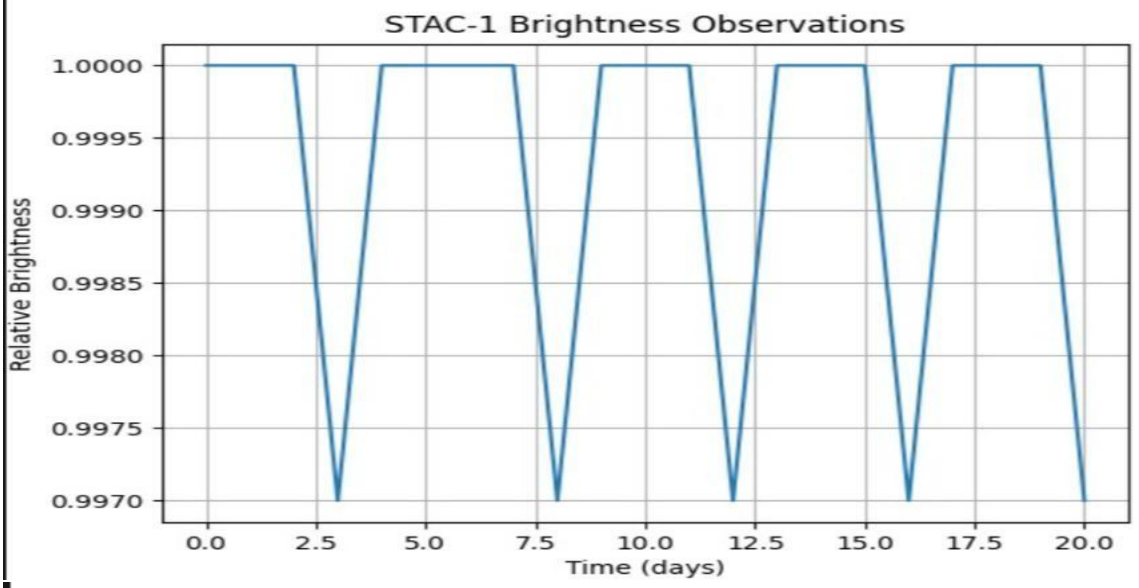
I. DETECTION AND IDENTIFICATION

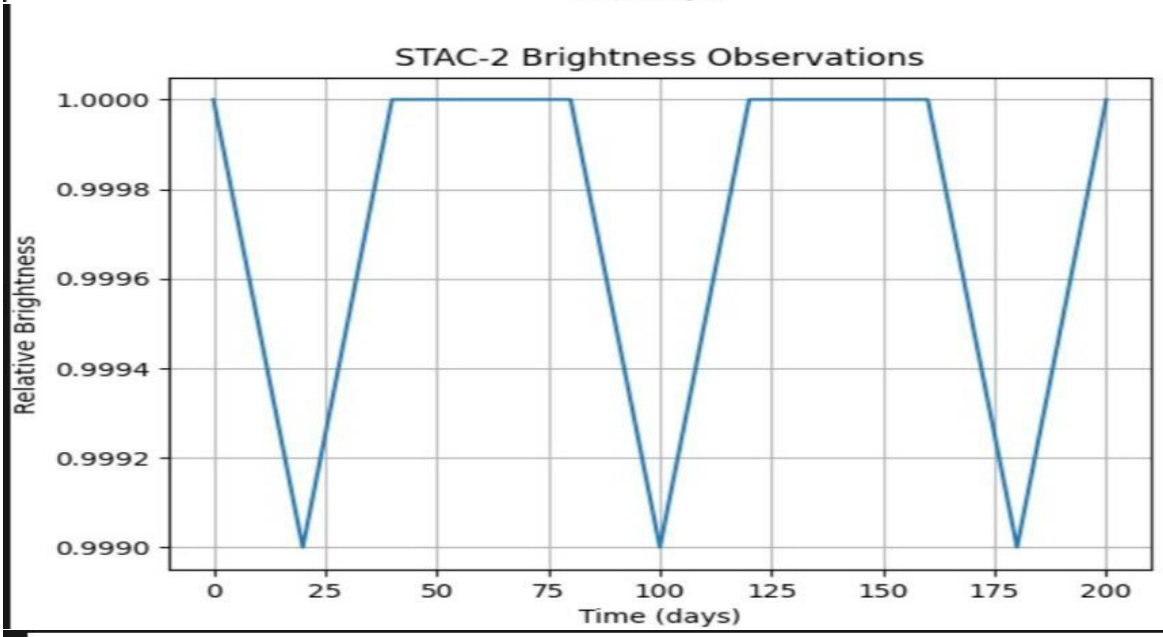
We looked at the brightness data for the three stars. A periodic dip in brightness means a planet is passing in front of the star (a transit).

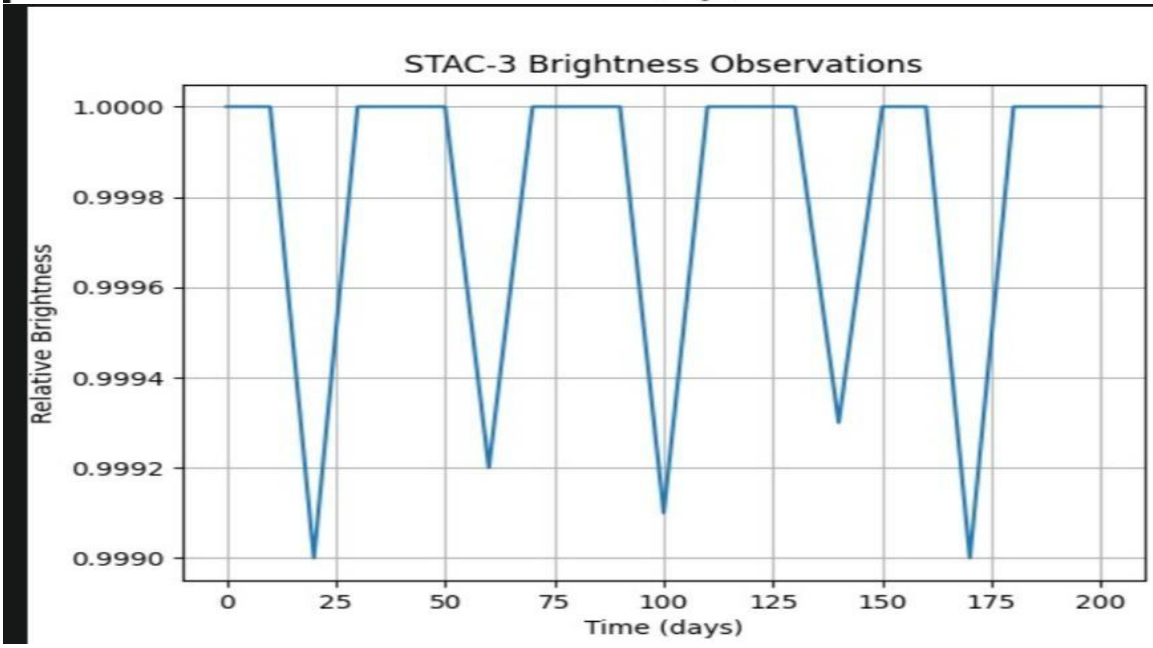
STAC-1 has regular dips in brightness. The transit period is about 4.25 days and the transit depth is 0.003. This indicates a transiting planet (Planet-1).

For **Planet-1**, applying the calculation to a transit depth of 0.003 yields an absolute radius of 38,121.49 km. Converting this metric relative to Earth yields approximately 5.98 Earth Radii ($R\oplus$), classifying Planet-1 as a Neptune-sized gas giant.

For **Planet-2**, the observed transit depth is 0.001. The calculated planetary radius is 17,607.56 km, which corresponds to 2.76 Earth Radii ($R\oplus$). Despite being smaller, it also falls within the broader Neptune-sized classification.


STAC-1 Brightness Observations
Relative Brightness
1.0000
0.9995
0.9990
0.9985
0.9980
0.9975
0.9970
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (days)


STAC-2 Brightness Observations
Relative Brightness
1.0000
0.9998
0.9996
0.9994
0.9992
0.9990
0
25
50
75
100
125
150
175
200
Time (days)


STAC-3 Brightness Observations
Relative Brightness
1.0000
0.9998
0.9996
0.9994
0.9992
0.9990
0
25
50
75
100
125
150
175
200
Time (days)

III. CALCULATIONS OF ORBITAL DISTANCES

Orbital distances (semi-major axis, a) were derived from the established transit periods (P) utilizing Kepler's Third Law. For a star with mass M in solar masses, the formula relates the period in Earth years to the distance in Astronomical Units (AU): $a\ (AU) = [\ (P_{days} / 365.25)^2 \times M\ ]^{1/3}$

For **Planet-1** (host star mass 1.0 M⊙), the 4.25-day period yields an orbital distance of 0.0513 AU, equating to 7,681,818.89 km from its host star.

For **Planet-2** (host star mass 0.8 M⊙), the extended 80.0day period results in a larger orbital distance of 0.3373 AU, translating to 50,461,648.11 km.

TABLE I. FINAL EXOPLANET REPORT SUMMARY

Property	Planet-1 (STAC-1)	Planet-2 (STAC-2)
Average Period (days)	4.25	80.0
Transit Depth	0.003	0.001
Radius (km)	38,121.49	17,607.56
Radius (Earth Radii)	5.98	2.76
Planet Type	Neptune-sized	Neptune-sized
Orbital Distance (AU)	0.0513	0.3373
Orbital Distance (km)	7,681,818.89	50,461,648.11
Habitable Zone Distance(AU)	1.0	0.63

IV. COMPARISON & IDENTIFICATION

We calculated the habitable zone distance using $d_{HZ} \approx \sqrt{(L/L\odot)}$ AU. For STAC-1, it is 1.0 AU. For STAC-2, it is $\sqrt{(0.40)} \approx 0.63$ AU.

Comparing the two, **Planet-2** is the most promising candidate. Planet-1 is at 0.0513 AU, which is way too close compared to its habitable zone of 1.0 AU. Planet2 is at 0.3373 AU, which is much closer to its habitable zone of 0.63 AU.

V. REASONING AND CONCLUSIONS

We choose Planet-2 as the best candidate based on these three points from our calculations:

1. **Distance to Habitable Zone:** Planet-2's orbit of 0.337 AU is reasonably close to its star's estimated habitable zone (0.63 AU). Planet-1 is far too close to its star.

2. **Temperature and Radiation:** Because Planet-1 is so close to its star, it receives too much heat, making liquid water impossible. Planet-2 is further away, making it a safer environment.

3. **Size:** Planet-2 is smaller (2.76 Earth radii) than Planet-1 (5.98 Earth radii). Smaller planets have a better chance of having a usable surface instead of just being huge gas planets.

Additional Observation: To check if Planet-2 can actually support life, we should look at the starlight passing through the planet's atmosphere when it transits. By studying the colors of this light, we can see if there is water vapor or oxygen present.

THANK YOU